Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details · Project: Ananta – Adaptive AI Tutor

a. **Team name:**
b. **Team leader name:**
c. **Problem Statement:**

Brief about the idea

A voice-first AI tutor that checks how a child thinks, not just what they answer.

The problem

- Children often don't ask in class: embarrassment, not knowing what to ask, fear of looking slow.
- A teacher cannot give every child unlimited, individual explanations.
- Quiz software only marks right or wrong, so a right answer from a wrong method counts as mastery.

The idea

- A judgment-free voice tutor teaches from the child's own textbook, line by line on a chalkboard.
- After each answer it diagnoses the reasoning, names the likely misconception, and re-teaches a different way.
- Mastery needs evidence of understanding, not one lucky answer.

Where it stands

- Working prototype, deployed on Google Cloud Run, built on Gemini.
- Textbook PDF in, structured curriculum out (concepts, prerequisites, misconceptions).
- Parents see the exact question and answer behind every claim. Built to work alongside teachers, not replace them.

Central principle: optimise for “the child understands it and can apply it”, not for “the AI answered the question”.

Opportunities

Most tutors answer questions. Ananta tests whether the child understands the answer.

How different is it from existing ideas?

- Chatbot tutors respond to questions; quiz apps check answers. Neither questions a correct answer.
- Ananta asks the child to explain, then classifies the reasoning against a closed catalogue of misconceptions taken from the textbook.
- A plain prompt to Gemini scored 9/10 on our hidden-misconception cases; the Ananta assessor scored 10/10 (small test, see slide 12).

How will it solve the problem?

- Private, unlimited, judgment-free asking by voice. One tap says “I don't get it”, “Another way”, “Too fast” or “I guessed”.
- Loop: teach, ask, diagnose, adapt, re-assess, record. The tutor switches representation (real-world, shape, 3D, chalkboard) using what has worked for this child.
- Never gives up: if a child is stuck it pivots to a worked example.

USP of the proposed solution

- Evidence-gated mastery: Bayesian Knowledge Tracing, plus at least two demonstrations of understanding.
- A deterministic teaching-plan compiler decides what to teach next, so planning is auditable, not hallucinated.
- Parent “See why” replay: the exact question, the child's own words, and how it was diagnosed.

List of features offered by the solution

Eight capabilities, each one visible in the live prototype.

Voice-guided chalkboard lesson

Gemini Live teaches by voice while one board fills line by line with chalk marks. Questions are written on the board first.

Reasoning diagnosis

Every answer is classed: clear reasoning, right answer / broken method, wrong, too thin, or none. Runs async, so the voice never waits.

Misconception ledger

Suspected, then confirmed after two independent observations, then resolved only by a sound transfer answer.

Adaptive re-teaching

Representation order ranked per child from what worked before: real-world, shape, 3D or chalkboard. Pace: slow, steady, quick.

Evidence-gated mastery

Bayesian Knowledge Tracing. Mastery needs pKnown of at least 0.80 and two level-3 demonstrations. A bare quiz click is capped at “recognised”.

Textbook PDF to curriculum

Upload via Files API. Gemini extracts concepts, prerequisites, worked examples and known misconceptions. Admin library included.

Quality-gated content

Lints, then an independent critic model, then repair. Anything that still fails is quarantined and never shown to a child.

Parent view

Plain-language report, misconception ledger, home activity, and “See why” evidence replay. Each sentence is checked against the child's real answers.

Process flow diagram or Use-case diagram

Content is prepared once; every live answer is diagnosed and fed back.

A. Before the lesson (once per textbook)

Upload textbook PDF + syllabus
Admin library

→

Gemini extracts curriculum
Concepts, prerequisites, misconceptions

→

Generate lessons + visuals
Chalkboard script, diagrams, photos

→

Critic review + lints
Fail = quarantine, never served

B. During the lesson (every answer)

Compile teaching plan
Deterministic rules from the child's history

→

Tutor teaches by voice
Board fills line by line

→

Child answers aloud
Or taps: not sure, another way

→

Diagnose reasoning
Gemini Flash, closed catalogue

→

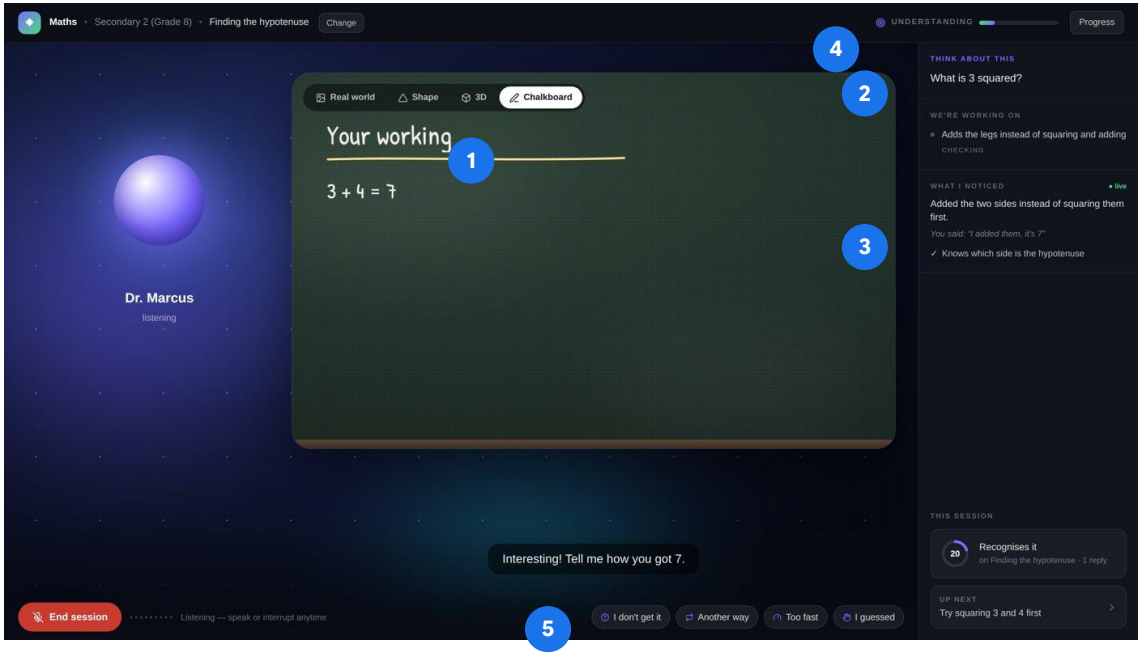
Update mastery + ledger
BKT, evidence event saved

Loop: next move chosen from the evidence: probe a prerequisite, re-teach with a different representation, ask a transfer question, or advance. Then back to the plan.

C. After the session A profiler (Gemini Flash) proposes claims about the child. Deterministic code accepts only claims backed by the child's own evidence. The Parent View shows them with a “See why” replay.

Wireframes/Mock diagrams of the proposed solution (optional)

The learning screen: what the child sees while the tutor diagnoses.



1 Chalkboard
Tutor writes while it speaks. Tabs switch representation: Real world, Shape, 3D, Chalkboard.

2 Think about this
Questions are asked on the right, answered by voice or tap.

3 We're working on
Live misconception card: confirmed, probing or resolved.

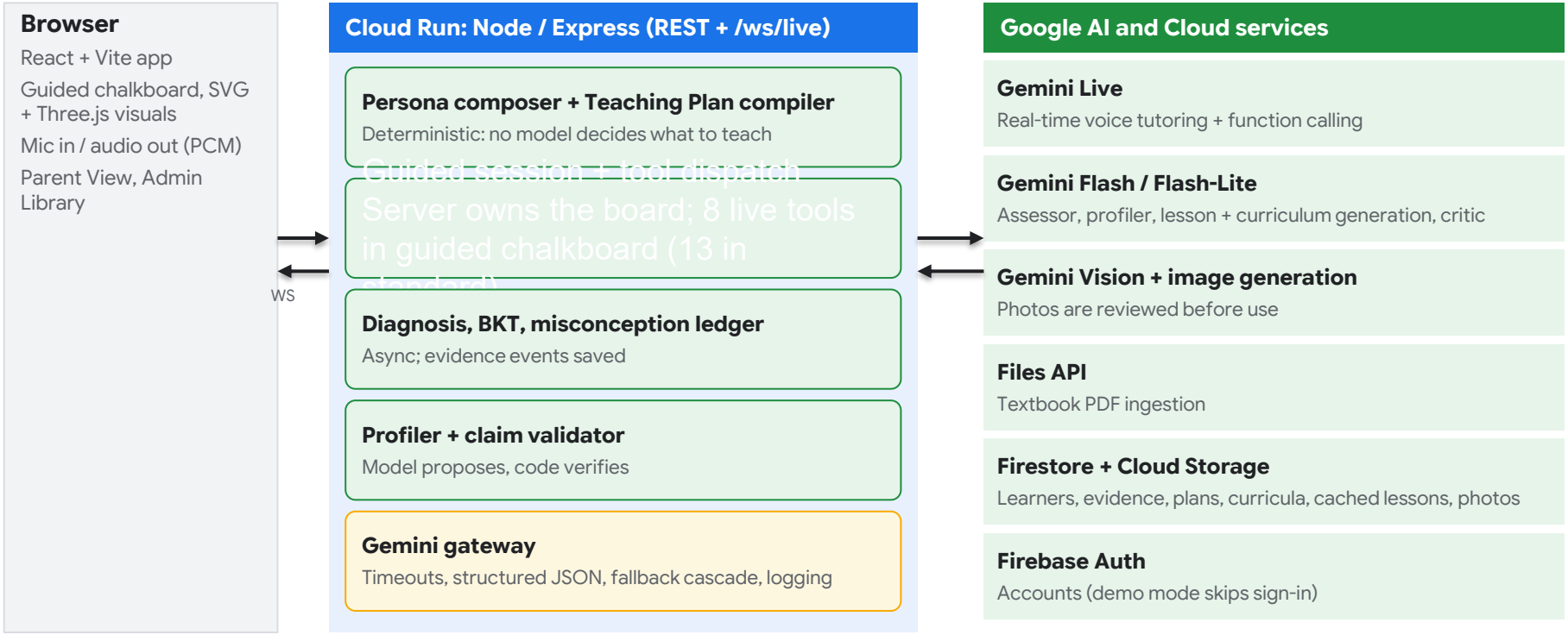
4 Understanding meter
Mastery shown as evidence-based levels, not a score.

5 Confusion buttons
I don't get it, Another way, Too fast, I guessed.

Parent Portal and Admin Library are separate screens reached from the start page.

Architecture diagram of the proposed solution

Gemini does the language work; deterministic code makes the decisions.



Technologies to be used in the solution

Google Cloud AI at every layer, in the roles each is best at.

Gemini 3.8 Live	Real-time two-way voice, voice activity detection, 8 function-calling tools in default guided chalkboard (13 in standard)
Gemini 3.8 Flash	Textbook ingestion (60-page chunks via Files API), lesson generation, session profiler, parent narrative, independent critic
Gemini 3.6 Flash	Per-answer reasoning diagnosis with 6.5s budget (measured p50: 3.15s, p95: 4.82s)
Gemini 3.1 Flash-Lite	Fallback for diagnosis and generation (resilience)
Gemini Vision / image	Generates real-world photos; the critic inspects each one before it can be served
Structured outputs	JSON schema on every model call; invalid output is rejected or repaired
Cloud Run	REST + WebSocket proxy + built app on one service; 3600 s timeout, session affinity, scale to zero
Firestore, Storage, Auth	Learner data is server-only (security rules deny client access); photos in Cloud Storage

Application: TypeScript, Node 22, Express, WebSocket, React, Vite, SVG and Three.js visuals, Docker. Built directly on the Google Gen AI SDK; agentic by design (tool calling, shared state in Firestore, multi-model orchestration).

Estimated implementation cost (optional)

Cost is designed in; a measured per-student figure comes from the first billing data.

Scale to zero

Cloud Run charges only while sessions run. Cold start on first request is the trade-off.

Generate once, serve many

Lessons, board pictures and photos are made at upload time, cached in Firestore / Storage and reused by every child.

Cheap model on the hot path

Diagnosis uses a fast Flash model; the profiler runs once per session, not per answer.

Gate before spend

Failed content is quarantined, not regenerated in a live session; repeat photo requests hit the cache.

What we know today

A long textbook is a paid, one-off generation job (several model calls per concept: draft, critic, repair).

Gemini free tier was used for the prototype and has rate limits.

To do: record Cloud Run + Gemini spend over a pilot and report cost per student-hour.

Snapshots of the prototype

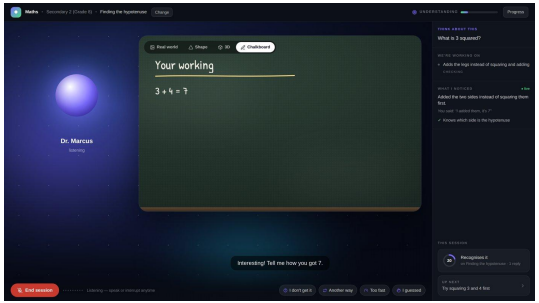
Real screens from the running app.



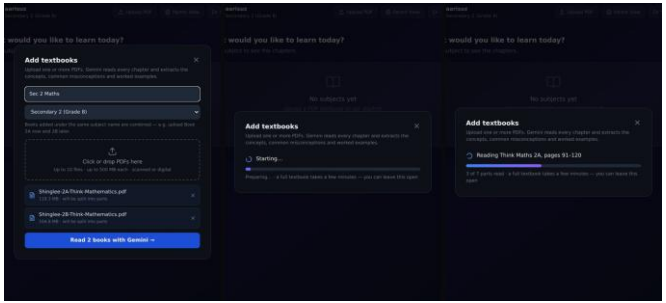
Real-world view: the child's ladder problem, with the tutor's question



Chalkboard view: worked steps written as the tutor speaks



Diagnosis in view: “3 + 4 = 7” flagged as adding instead of squaring



Admin: upload textbook PDFs, Gemini reads them into a curriculum

Prototype Performance report/Benchmarking

Catches a right answer reached by a broken method (10 cases)



96%
exact class, 43/45

0/10
false alarms on sound reasoning

10/10
right misconception named

2/2
prompt-injection cases held

Honest numbers
Latency per diagnosis in this run: p50 3.15 s, p95 4.82 s, max 5.79 s, mean 3.32 s (n=45) — well under our 6.5s budget!
Flags are 100% precise: 10 of 10 flags were right (plain prompt: 10 of 11, 91%).

How to read this
Run on gemini-3.6-flash (2026-10-17). 45 hand-written cases across 5 concepts with Wilson 95% intervals.
Only 10 hidden-misconception cases: 95% CI for 10/10 is 72–100%. Tests the single-answer assessor.
Also verified: app shell loads in 102 ms; learner data server-only; test suite passing for quality

Additional Details/Future Development (if any)

Next: prove learning gains with real children and teachers.

Next 3 months

- Pilot with a teacher and a small class; measure learning gain against a control.
- Independent annotators re-label the evaluation set; grow it past 40 cases per concept.
- Bring diagnosis latency down toward the 4 s target.
- Spaced-review scheduling and decay of old mastery.
- Teacher view: class-level misconception heatmap.

Scale

- Any curriculum: upload a textbook PDF and syllabus. Boards in the library: IGCSE, Cambridge, CBSE, US Common Core.
- More subjects beyond Mathematics and Biology.
- Preferred language captured at onboarding; broaden voice languages.
- Supports SDG 4, quality education, by making individual explanation available outside class hours.

Limits to know about

- Voice needs a microphone, a quiet room, and Chrome or Firefox.
- Pause briefly between sentences (voice activity detection).
- Gemini free tier has rate limits; if replies slow, wait a few seconds.
- First load can take 10–20 s (Cloud Run cold start).
- Photo generation may fall back to diagrams if image quota runs out.
- Demo learners are simulated and labelled so.

Provide links to your:

- GitHub Public Repository:
- Demo Video Link (3 Minutes):
- Final Product Link:

For evaluators: test in under a minute

1. Open the product link. No sign-in is needed (demo mode).
2. Click "Maya (Simulated)" or "Leo (Simulated)" (Grade 4 Mathematics, Money and decimal place value). Notice Ananta opens with a place-value diagram for Maya, but with a worked example and avoids diagrams for Leo.
3. Try asking "Can you show me a picture?" or answering "10 coins" to see the diagnostic assessment and BKT update in real time.
4. Open the Parent Portal from the start page to see Ananta's plan for the next lesson and click "See

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Thank you

